

Tutorial - 3

2> 390 errors in 520 page
 $\# \text{ errors / page} = \frac{390}{520} = 0.75.$

$\lambda = 0.75.$
 $X: \# \text{ errors / page}$
 $X \sim P(\lambda) \quad \lambda = 0.75$

$X_1, X_2, X_3, \dots$
 $X_i \sim P(\lambda)$
 $P(X=0) = \left(\frac{0.75}{e}\right)^5$

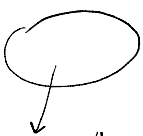
11> $N = 100$ # typos
 $N_p = 10$ # defective typos
 $\text{Sample size } m = 4.$
 $P = \frac{N_p}{N} = \frac{1}{10}$

$X: \# \text{ defective typos in the sample of size 4.}$

a) $P(X=1) = \frac{\binom{10}{1} \cdot \binom{90}{3}}{\binom{100}{4}}$

b) $E(X) = mp = 4 \times 0.1$

c) $\text{Var}(X) = mpq \frac{N-m}{N-1} \quad (q = 1-p)$

3> 
 $n = 10^4$
 $P: \text{probability of having an accident.}$
 $P = 0.1\% = 0.001$

$P(X \leq 5) = ?$

$np = 10^4 \times 0.001 = 10$

Using Binomial to Poisson approximation
 $P(X \leq 5)$

$\lambda = 10, P(X=2)$

Using binomial

$$P(X \leq 5)$$

$$= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= e^{-10} \left(\frac{10^0}{0!} + \frac{10}{1!} + \frac{10^2}{2!} + \frac{10^3}{3!} + \frac{10^4}{4!} + \frac{10^5}{5!} \right)$$

4

P : probability of suffering due injection

$$P = 0.001$$

$$n = 2000$$

$$np = 2000 \times 0.001 = 2 = \lambda$$

$$a) P(X=3) = e^{-2} \frac{2^3}{3!}$$

$$b) P(X \geq 2) = 1 - P(X=0) - P(X=1) - P(X=2)$$

5

$n = 10$ blades.

p : prob that blade being defective

$$p = \frac{1}{5 \times 100} = 0.002$$

$$n = 10 \times 0.002 = 0.02 = \lambda$$

$$a) P(X=0) \times 10^5 \rightarrow \left(\frac{e^{-\lambda} \lambda^x}{x!} \right) \times 10^5 \rightarrow \left(\frac{e^{-0.02} (0.02)^0}{0!} \right) \times 10^5 \rightarrow 0.98 \times 10^5$$

~~10~~ a) $P(X=0)$
 $= e^{-0.02} \times 10^5$

10⁵ packets containing no defective blades.

b) $P(X=1) \times 10^5$

c) $P(X=2) \times 10^5$

6

$n = 120$ $p = \frac{5}{100}$

$np = 1$

$P(X=0) = e^{-1}$

7

X : # failure before 1st success

defective - success
good - failure

$\begin{array}{ccccccc} F & & & & S \\ 0 & \dots & 0 & 0 \\ \hline 8 & & & 9 \end{array}$

$P(X=8) = \cancel{0.01} (0.99)^8 (0.01)$

8

$p = \frac{12}{100}$

$\begin{array}{ccccccc} F & F & F & F & S \\ 0 & 0 & 0 & 0 & 0 \\ \hline & & & & 5 \end{array}$

$P(X=4)$

9

10 gaskets
4 were defective

$Np = 4$
 $\frac{NP}{n} = \frac{4}{10}$

$N = 10$

$\rightarrow m = 3$

X : # defective in $m=3$ gaskets

$P(X=2)$

$$NP = 4$$

$$p = \frac{NP}{N} = \frac{4}{10}$$

$$6 \mid 4$$

$$P(X=2)$$

$$= \frac{\binom{4}{2} \binom{6}{1}}{\binom{10}{3}}$$